

Verdant Coil: Development Roadmap (Refined)

Overview

This phased roadmap breaks down the development of *Verdant Coil* into focused stages. Each phase results in testable, functional modules. Prioritization balances rapid prototyping with future extensibility.

Phase 1: Core Gameplay Prototype (MVP)

Goal: Create a local prototype of the full player loop using JSON and placeholder graphics.

1.1 Crawler Movement & Control

- Grid-based movement with smooth tile transitions
- Contextual interactions (hazard triggers, wall blocks)

1.2 Simple Upgrade System

- 3 toggleable upgrades:
 - Hardened Skin
 - Acid Sac
 - Directional Memory
- System to check if player has upgrade (no economy yet)

1.3 Tile System & Modules

- Placeable grid with limited module types:
 - Flesh Tile (walkable)
 - Acid Pool (damage)
 - Sticky Web (slow)
 - Digest Wall (destructible)
 - Heartroot (goal)

1.4 Builder Mode (Local)

- Drag-and-place interface on 2D grid
- Biomass budget enforced
- Internal level validation: must complete to publish

1.5 Save/Load System

- Levels and player data saved as JSON
 - Load/play existing levels from local files
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Phase 2: Greybox Playtest & Feedback Pass

Goal: Tighten core feel through internal testing before adding meta systems.

2.1 UX Polishing

- Undo/redo system in builder
- Basic crawler animations (squash/stretch)
- Damage and slow visual feedback

2.2 Level Design QA Tools

- Path validator
- Highlight required upgrades

2.3 Local Profile Tracking

- Store crawler progress and unlocked upgrades in local file
 - Replay last 5 played levels
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Phase 3: Local Sharing System

Goal: Allow asynchronous sharing without full backend.

3.1 Level Browser

- Browse local levels with metadata (author, rating, play count)
- Star rating system (local-only)

3.2 Local Publishing Format

- Publish = move to /Published folder with metadata
 - Import levels from others (zip or folder)
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Phase 4: Backend Infrastructure (Optional Post-MVP)

Goal: Introduce real sharing and persistence across devices.

4.1 Auth & Sync

- UUID-based player identity
- Level sync, resource sync

4.2 Play Tracking

- Server logs completions and play count
- Reward creators based on completions

4.3 Anti-Abuse Logic

- Prevent self-farming via IP/session check
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Phase 5: Meta Systems & Cosmetic Polish

Goal: Deepen progression and enhance user engagement.

5.1 Economy System

- Earn Nutrients by completing levels
- Earn Sporeprint when your level is completed (stubbed if no backend)
- Spend resources to unlock upgrades or tiles

5.2 New Biomes and Tiles

- Add 2nd theme (e.g. Techno-Fungus)
- New module types (e.g. Pulse Gate, Trigger Node)

5.3 Dashboards & Discovery

- Show stats: total plays, ratings, most successful Coils
 - Featured/rotating challenge system
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Suggested Milestone Releases



Milestone 1: Local MVP (End of Phase 1)

Playable crawler + builder + tile validation + local save/load



Milestone 2: Local Sharing (End of Phase 3)

Play and rate others' levels via import/export



Milestone 3: Online Sharing (End of Phase 4)

Persistent sharing with server reward tracking

Milestone 4: Progression Loop + Polish (End of Phase 5)

Resource economy, cosmetics, dashboards, and events